

回顧 集合 $\mathcal{P}$:

- (集合運算) $\bigcap : P(\mathcal{P}) \rightarrow \mathcal{P}$

- (算子) $\cdot^c : \mathcal{P} \rightarrow \mathcal{P}$

滿足

- (冪等律) $\forall A \in \mathcal{P} \quad \bigcap \{A\} = A$
 $\bigcup \{A\} = A$

- (結合律) $\forall C$ 劃分 $\mathcal{A} \subseteq \mathcal{P} \quad \bigcap \mathcal{A} = \bigcap \{ \bigcap \mathcal{C} : \mathcal{C} \in C \}$
 $\bigcup \mathcal{A} = \bigcup \{ \bigcup \mathcal{C} : \mathcal{C} \in C \}$

- (分配律) $\forall C \subseteq P(\mathcal{P}) \quad \bigcap \bigcup_{\mathcal{C} \in C} \mathcal{C} = \bigcup_{\mathbf{x} \in \prod_{\mathcal{C} \in C} \mathcal{C}} \bigcap X_{\mathcal{C}}$
 $\bigcup \bigcap_{\mathcal{C} \in C} \mathcal{C} = \bigcap_{\mathbf{x} \in \prod_{\mathcal{C} \in C} \mathcal{C}} \bigcup X_{\mathcal{C}}$

- (單位元) $\exists \frac{U}{\emptyset} \in \mathcal{P} \text{ st } \forall A \in \mathcal{P} \quad \begin{matrix} A \cap U = A = ' U \cap A \\ A \cup \emptyset = A = ' \emptyset \cup A \end{matrix}$

- (de Morgan) $\forall \mathcal{A} \subseteq \mathcal{P} \quad \begin{matrix} (\bigcap_{A \in \mathcal{A}} A)^c = \bigcup_{A \in \mathcal{A}} A^c \\ (\bigcup_{A \in \mathcal{A}} A)^c = \bigcap_{A \in \mathcal{A}} A^c \end{matrix}$

- (互補律) $\forall A \in \mathcal{P} \quad \begin{matrix} A \cap A^c = \emptyset = ' A^c \cap A \\ A \cup A^c = U = ' A^c \cup A \end{matrix}$

稱為 *集合系統

顯然 $\forall$ 集合 $X \quad (P(X), \bigcap, \bigcup)$ 是 *集合系統 定義確實以集合觀點推廣 :

集合 $\Rightarrow$ 預集合

已知 $\mathcal{B} \subseteq \mathcal{P}$ 是解析族 $(\sum \mathcal{B} = U) \Rightarrow$

$$E_{\langle \mathcal{B} \rangle} = \mathcal{B} \quad (\langle \mathcal{B} \rangle = \{ \sum \mathcal{F} : \mathcal{F} \subseteq \mathcal{B} \})$$

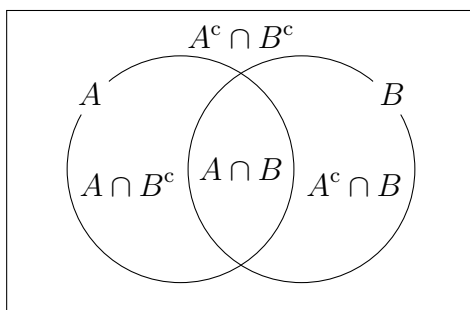
且有自然同構

$$\sum \mathcal{F} \rightarrow \mathcal{F} : \langle \mathcal{B} \rangle \rightarrow P(\mathcal{B})$$

因此我們的策略為 :

$$\forall \mathcal{A} \subseteq \mathcal{P} \quad \text{找到解析族 } \mathcal{B} \subseteq \langle \mathcal{A} \rangle \text{ st } E_{\langle \mathcal{A} \rangle} = \mathcal{B}$$

注意到



因此我們猜

$$E_{\langle \mathcal{A} \rangle} = \{E_{\mathcal{F}} := \bigcap_{X \in \mathcal{A}} \sigma_{\mathcal{F}}(X) : \mathcal{F} \subseteq \mathcal{A}\}$$

其中 $\sigma_{\mathcal{F}}(X) := \begin{cases} X \in \mathcal{F} \Rightarrow X \\ X \notin \mathcal{F} \Rightarrow X^c \end{cases} :$

1. $E_{\mathcal{F}} \cap E_{\mathcal{G}} = \delta_{\mathcal{F}\mathcal{G}} :$

$$\mathcal{F} \neq \mathcal{G} \Rightarrow \exists C \in \frac{\mathcal{F}}{\mathcal{G}} \text{ st } C \notin \frac{\mathcal{G}}{\mathcal{F}} \Rightarrow E_{\mathcal{F}} \cap E_{\mathcal{G}} \subseteq \sigma_{\mathcal{F}}(C) \cap \sigma_{\mathcal{G}}(C) = \emptyset$$

2. (康托爾對角線) $\bigcup_{\mathcal{F} \subseteq \mathcal{A}} E_{\mathcal{F}} = \mathbf{U} :$

$$\bigcup_{\mathcal{F} \subseteq \mathcal{A}} \bigcap_{X \in \mathcal{A}} \sigma_{\mathcal{F}}(X) = \bigcap_{\mathbf{X} \in \mathcal{A}^{\mathbf{P}(\mathcal{A})}} \left(\bigcup_{\mathcal{F} \subseteq \mathcal{A}} \sigma_{\mathcal{F}}(X_{\mathcal{F}}) \supseteq \sigma_{\mathcal{C}}(X_{\mathcal{C}}) + \sigma_{\mathcal{D}}(X_{\mathcal{D}}) = \mathbf{U} \right)$$

其中 $\mathcal{C} := \{X_{\mathcal{F}} : X_{\mathcal{F}} \notin \mathcal{F}\}$ st

- $X_{\mathcal{C}} \in \mathcal{C}$ ($X_{\mathcal{C}} \notin \mathcal{C} \Rightarrow X_{\mathcal{C}} \in \mathcal{C}$)
- $\exists X_{\mathcal{D}} \notin \mathcal{D}$ st $X_{\mathcal{C}} = X_{\mathcal{D}} \Rightarrow \mathcal{C} \neq \mathcal{D}$ ($\mathcal{C} = \mathcal{D} \Rightarrow X_{\mathcal{D}} \in \mathcal{D}$)

3. $\forall A \in \mathcal{A} \quad A \in \langle E_{\langle \mathcal{A} \rangle} \rangle :$

$$A \cap \mathbf{U} = \sum_{\mathcal{F} \subseteq \mathcal{A}} A \cap E_{\mathcal{F}} = \sum_{A \in \mathcal{F} \subseteq \mathcal{A}} E_{\mathcal{F}} \quad (A \notin \mathcal{F} \Rightarrow A \cap E_{\mathcal{F}} \subseteq A \cap \sigma_{\mathcal{F}}(A) = \emptyset)$$

代 $\mathcal{A} = \mathcal{P}$ 得到 $\mathcal{P} \simeq \mathbf{P}(E_{\mathcal{P}})$ ，定理確實找出相應的元素觀點：

集合 $\Leftarrow$ 預集合

以惠勒式的科普話語，即

「元素組成集合，集合分解元素」